DATA WAREHOUSING — LABORATORY ACTIVITY 4

Technical Data Dictionary & Metadata Architecture Document

Performance Tuning, Data Governance, Schema Validation & Departmental Data Mart Creation

Group Project:	Group 11 — Streaming Media User Subscription & Watch-Time Analytics Warehouse
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1. Executive Architectural Overview & Schema Blueprint

This technical data dictionary details the comprehensive enterprise metadata architecture for Group 11's Streaming Media Data Warehouse. Designed to solve peak-hour analytical query bottlenecks and isolate business unit workloads without compromising underlying data security, the platform is structured across four high-performance tiers:

• **Tier 1: Core Relational Star Schema (warehouse.db):** Separates operational transactional records into central fact table (watch_logs) surrounded by high-cardinality dimension tables (content_catalog, user_subscriptions). Optimized with custom single-column and compound B-Tree indexes, WAL journaling, and 64MB caching.

• **Tier 2: Governed Relational Schema (warehouse_governed.db):** Enforces strict table-level DDL invariants including NOT NULL constraints, positive numeric CHECK rules on subscription fees and watch durations, bounded completion ratios (0.0 to 1.0), and cascading/restrict foreign keys.

• **Tier 3: Isolated Departmental Data Mart (viewer_watchtime_mart.db):** A downstream, isolated analytical repository dedicated to the Content Programming & Audience Engagement Department. Contains high-speed aggregated summary tables materializing monthly genre trends, title engagement tiers, user viewing streak habituation, and daily traffic metrics.

• **Tier 4: Governed Document Store (streaming_governance_dw in MongoDB):** Captures semi-structured client device telemetry and unstructured viewer feedback, fortified by strict $jsonSchema validators that reject malformed or non-compliant payloads at the database engine level.

2. Comprehensive Entity Data Dictionaries

2.1 Table / Entity: content_catalog (Core Dimension Table (SQLite))

Target Database:	warehouse.db	**Key Topology:**	PK: show_id
Business Purpose:	Stores cleansed, deduplicated video titles, cast, directors, release metadata, and standardized runtimes for streaming titles.	**Foreign Keys:**	None
Indexes / Rules:	sqlite_autoindex_content_catalog_1 (PK), idx_content_catalog_type, idx_content_catalog_release_year, idx_catalog_type_year	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
show_id	TEXT	PK	NOT NULL	s\d+ format	None	Unique alphanumeric identifier for streaming title	netflix_titles.csv: show_id
type	TEXT	None	NOT NULL	'Movie' | 'TV Show'	None	Primary content classification category	netflix_titles.csv: type
title	TEXT	None	NOT NULL	Trimmed string > 0	None	Full commercial display title of the media asset	netflix_titles.csv: title
director	TEXT	None	NULL	Free text	None	Directorial credits; comma-separated for co-directors	netflix_titles.csv: director
cast	TEXT	None	NULL	Free text	None	Key billing cast members; comma-separated	netflix_titles.csv: cast
country	TEXT	None	NULL	ISO country list	None	Countries of production origin	netflix_titles.csv: country
date_added	TEXT	None	NULL	YYYY-MM-DD	None	Standardized date asset became available on service	netflix_titles.csv: date_added
release_year	INTEGER	None	NOT NULL	1900 - 2030	None	Original theatrical or television release year	netflix_titles.csv: release_year
rating	TEXT	None	NULL	MPAA / TV Rating	None	Age-appropriateness certification (e.g., TV-MA, PG-13)	netflix_titles.csv: rating
duration_raw	TEXT	None	NULL	Free text	None	Original unparsed string from source ingestion	netflix_titles.csv: duration
duration_num	INTEGER	None	NULL	> 0	Median impute	Extracted numeric value of duration (minutes or seasons)	ETL clean_duration() regex
duration_unit	TEXT	None	NULL	'Min' | 'Season(s)'	Type-based	Unit descriptor corresponding to duration_num	ETL clean_duration() regex
listed_in	TEXT	None	NULL	Comma-separated	None	Hierarchical genre descriptors	netflix_titles.csv: listed_in
description	TEXT	None	NULL	Free text	None	Promotional synopsis and plot summary	netflix_titles.csv: description

2.2 Table / Entity: user_subscriptions (Core Dimension Table (SQLite))

Target Database:	warehouse.db	**Key Topology:**	PK: user_id
Business Purpose:	Maintains subscriber account status, plan tiers, billing amounts, and auto-renew renewal flags.	**Foreign Keys:**	None
Indexes / Rules:	sqlite_autoindex_user_subscriptions_1 (PK), idx_user_subscriptions_status, idx_user_subscriptions_plan, idx_user_subs_status_plan	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
user_id	INTEGER	PK	NOT NULL	>= 1000	None	Surrogate unique subscriber account identifier	Synthetic Generator (1001-1500)
signup_date	TEXT	None	NOT NULL	YYYY-MM-DD	None	Date of account registration and initial billing	Synthetic generator (2025-01-01+)
plan_type	TEXT	None	NOT NULL	'Basic' | 'Standard' | 'Premium'	None	Subscription tier determining concurrency and bitrate	Categorical choice [0.3, 0.5, 0.2]
monthly_fee	REAL	None	NOT NULL	> 0.00	None	Recurring monthly billing charge in USD	Plan mapping ($9.99, $15.49, $19.99)
auto_renew	INTEGER	None	NOT NULL	0 or 1	1	Binary flag indicating automated billing renewal enabled	Bernoulli sample [0.2, 0.8]
status	TEXT	None	NOT NULL	'Active' | 'Cancelled' | 'Expired'	None	Current lifecycle account status	Active if auto_renew else churn

2.3 Table / Entity: watch_logs (Core Fact Table (SQLite))

Target Database:	warehouse.db	**Key Topology:**	PK: log_id
Business Purpose:	Records individual content streaming events, session durations, and user completion ratios.	**Foreign Keys:**	user_id -> user_subscriptions(user_id), show_id -> content_catalog(show_id)
Indexes / Rules:	sqlite_autoindex_watch_logs_1 (PK), idx_watch_logs_user_id, idx_watch_logs_show_id, idx_watch_logs_watch_date, idx_watch_logs_user_date, idx_watch_logs_show_dur_comp, idx_watch_logs_date_comp	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
log_id	INTEGER	PK	NOT NULL	Auto-increment	None	Surrogate synthetic primary key for watch transaction	SQLite AUTOINCREMENT
user_id	INTEGER	FK	NOT NULL	>= 1000	None	Foreign key linking to subscriber account	FK user_subscriptions(user_id)
show_id	TEXT	FK	NOT NULL	s\d+ format	None	Foreign key referencing streaming media title	FK content_catalog(show_id)
watch_date	TEXT	None	NOT NULL	YYYY-MM-DD	None	Calendar date on which the streaming session took place	Synthetic session timestamp
watch_duration_minutes	INTEGER	None	NOT NULL	1 - 1440	None	Total active playback elapsed duration in minutes	Uniform integer [10, 180]
completion_percentage	REAL	None	NOT NULL	0.00 - 1.00	None	Ratio of playback progress relative to total runtime	Uniform float [0.10, 1.00]

2.4 Table / Entity: content_catalog_governed (Governed Dimension Table (SQLite))

Target Database:	warehouse_governed.db	**Key Topology:**	PK: show_id
Business Purpose:	Enterprise-governed dimension catalog enforcing strict relational CHECK constraints, type invariants, and release boundaries.	**Foreign Keys:**	None
Indexes / Rules:	PK (show_id), CHECK (type IN ('Movie','TV Show')), CHECK (release_year BETWEEN 1900 AND 2030), CHECK (duration_num > 0)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
show_id	TEXT	PK	NOT NULL	s\d+	None	Validated primary content identifier	Transferred from content_catalog
type	TEXT	None	NOT NULL	CHECK IN ('Movie', 'TV Show')	None	Strictly constrained content classification	Validated type column
title	TEXT	None	NOT NULL	CHECK length(trim(title)) > 0	None	Guaranteed non-empty media asset title	Validated title column
release_year	INTEGER	None	NOT NULL	CHECK release_year BETWEEN 1900 AND 2030	None	Chronologically bounded historical release year	Validated release_year
duration_num	INTEGER	None	NULL	CHECK duration_num > 0	None	Positively bounded duration count	Validated duration_num
duration_unit	TEXT	None	NULL	CHECK IN ('Min', 'Season', 'Seasons')	None	Standardized runtime unit descriptor	Validated duration_unit

2.5 Table / Entity: user_subscriptions_governed (Governed Dimension Table (SQLite))

Target Database:	warehouse_governed.db	**Key Topology:**	PK: user_id
Business Purpose:	Governed subscriber dimension enforcing positive pricing bounds, valid tier enums, and ISO signup date formats.	**Foreign Keys:**	None
Indexes / Rules:	PK (user_id), CHECK (user_id >= 1000), CHECK (plan_type IN ('Basic', 'Standard', 'Premium')), CHECK (monthly_fee > 0.0)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
user_id	INTEGER	PK	NOT NULL	CHECK user_id >= 1000	None	Positive integer unique subscriber ID	Transferred from user_subscriptions
signup_date	TEXT	None	NOT NULL	CHECK strftime ISO valid	None	ISO-8601 standardized signup date	Validated date format
plan_type	TEXT	None	NOT NULL	CHECK IN ('Basic', 'Standard', 'Premium')	None	Restricted authorized subscription tier enum	Validated tier string
monthly_fee	REAL	None	NOT NULL	CHECK monthly_fee > 0.0	None	Guaranteed positive recurring subscription fee	Validated billing decimal
auto_renew	INTEGER	None	NOT NULL	CHECK IN (0, 1)	1	Validated binary renewal flag	Validated integer flag
status	TEXT	None	NOT NULL	CHECK IN ('Active', 'Cancelled', 'Expired')	None	Validated subscription lifecycle status	Validated status enum

2.6 Table / Entity: watch_logs_governed (Governed Fact Table (SQLite))

Target Database:	warehouse_governed.db	**Key Topology:**	PK: log_id
Business Purpose:	Governed streaming fact table enforcing referential integrity with cascading/restrict rules and bounded playback metrics.	**Foreign Keys:**	user_id -> user_subscriptions_governed(user_id) ON DELETE CASCADE, show_id -> content_catalog_governed(show_id) ON DELETE RESTRICT
Indexes / Rules:	PK (log_id), FK user_id, FK show_id, CHECK (duration BETWEEN 1 AND 1440), CHECK (completion BETWEEN 0.0 AND 1.0)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
log_id	INTEGER	PK	NOT NULL	AUTOINCREMENT	None	Primary transaction record ID	Transferred from watch_logs
user_id	INTEGER	FK	NOT NULL	REFERENCES user_subscriptions_governed	None	Strict referential integrity child foreign key	Foreign key constraint
show_id	TEXT	FK	NOT NULL	REFERENCES content_catalog_governed	None	Strict referential integrity child foreign key	Foreign key constraint
watch_date	TEXT	None	NOT NULL	CHECK strftime ISO valid	None	Validated session calendar date	Validated date format
watch_duration_minutes	INTEGER	None	NOT NULL	CHECK duration > 0 AND duration <= 1440	None	Guaranteed realistic playback session length	Validated duration
completion_percentage	REAL	None	NOT NULL	CHECK completion >= 0.0 AND completion <= 1.0	None	Strictly bounded completion ratio between 0% and 100%	Validated ratio

2.7 Table / Entity: monthly_genre_performance (Departmental Data Mart Table (SQLite))

Target Database:	viewer_watchtime_mart.db	**Key Topology:**	PK: Composite (year_month, genre)
Business Purpose:	Aggregated monthly performance metrics by genre, tracking viewer volume, watch time, completion rates, and attributed revenue.	**Foreign Keys:**	None (Isolated analytical mart)
Indexes / Rules:	idx_mart_genre_month ON (year_month, genre)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
year_month	TEXT	PK (Part)	NOT NULL	YYYY-MM	None	Reporting cohort calendar period (Year-Month)	strftime('%Y-%m', watch_date)
genre	TEXT	PK (Part)	NOT NULL	Genre String	None	Individual unnested content genre classification	json_each unnesting of cc.listed_in
total_streams	INTEGER	None	NOT NULL	>= 0	0	Total volume of streaming sessions initiated in period	COUNT(log_id)
unique_viewers	INTEGER	None	NOT NULL	>= 0	0	Count of distinct active subscribers consuming genre	COUNT(DISTINCT user_id)
total_watch_minutes	INTEGER	None	NOT NULL	>= 0	0	Sum of elapsed viewing minutes across genre sessions	SUM(watch_duration_minutes)
total_watch_hours	REAL	None	NOT NULL	>= 0.0	0.0	Sum of elapsed viewing hours rounded to 2 decimals	ROUND(total_watch_minutes / 60.0, 2)
avg_completion_pct	REAL	None	NOT NULL	0.0 - 100.0	0.0	Mean content completion percentage achieved	ROUND(AVG(completion_percentage)*100, 2)
attributed_revenue	REAL	None	NOT NULL	>= 0.0	0.0	Estimated attributed subscription revenue for genre	ROUND(SUM(monthly_fee), 2)
revenue_per_watch_hour	REAL	None	NOT NULL	>= 0.0	0.0	Monetization efficiency ratio (attributed revenue / hours)	ROUND(revenue / hours, 2)

2.8 Table / Entity: content_engagement_summary (Departmental Data Mart Table (SQLite))

Target Database:	viewer_watchtime_mart.db	**Key Topology:**	PK: show_id
Business Purpose:	Title-level engagement rollup classifying content into performance tiers based on cumulative watch hours and viewer reach.	**Foreign Keys:**	None (Isolated analytical mart)
Indexes / Rules:	idx_mart_content_views ON (total_watch_minutes DESC)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
show_id	TEXT	PK	NOT NULL	s\d+	None	Catalog identifier for the media asset	content_catalog: show_id
title	TEXT	None	NOT NULL	Trimmed string	None	Full commercial display title	content_catalog: title
content_type	TEXT	None	NOT NULL	'Movie' | 'TV Show'	None	Classification type	content_catalog: type
release_year	INTEGER	None	NOT NULL	1900 - 2030	None	Original release year	content_catalog: release_year
rating	TEXT	None	NULL	Rating string	None	Audience rating maturity classification	content_catalog: rating
total_streams	INTEGER	None	NOT NULL	>= 1	None	Lifetime stream count on platform	COUNT(log_id)
unique_viewers	INTEGER	None	NOT NULL	>= 1	None	Number of distinct viewers who engaged with title	COUNT(DISTINCT user_id)
total_watch_minutes	INTEGER	None	NOT NULL	>= 0	0	Cumulative playback minutes accumulated	SUM(watch_duration_minutes)
total_watch_hours	REAL	None	NOT NULL	>= 0.0	0.0	Cumulative playback hours accumulated	ROUND(total_watch_minutes / 60.0, 2)
avg_completion_pct	REAL	None	NOT NULL	0.0 - 100.0	0.0	Mean watch completion percentage	ROUND(AVG(completion_percentage)*100, 2)
engagement_tier	TEXT	None	NOT NULL	Tier Enum	None	Categorical rating ('Blockbuster Tier', 'High', 'Moderate', 'Niche')	CASE on total_watch_minutes threshold

2.9 Table / Entity: subscriber_viewing_habits (Departmental Data Mart Table (SQLite))

Target Database:	viewer_watchtime_mart.db	**Key Topology:**	PK: user_id
Business Purpose:	User-level behavioral profile detailing viewing streaks, watch intensity, viewer segment classification, and churn risk indicators.	**Foreign Keys:**	None (Isolated analytical mart)
Indexes / Rules:	idx_mart_user_segment ON (viewer_segment, retention_status)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
user_id	INTEGER	PK	NOT NULL	>= 1000	None	Surrogate user account identifier	user_subscriptions: user_id
plan_type	TEXT	None	NOT NULL	'Basic' | 'Standard' | 'Premium'	None	Active subscription tier	user_subscriptions: plan_type
account_status	TEXT	None	NOT NULL	'Active' | 'Cancelled' | 'Expired'	None	Current account lifecycle status	user_subscriptions: status
signup_date	TEXT	None	NOT NULL	YYYY-MM-DD	None	Registration cohort date	user_subscriptions: signup_date
monthly_fee	REAL	None	NOT NULL	> 0.0	None	Monthly subscription charge	user_subscriptions: monthly_fee
total_sessions_logged	INTEGER	None	NOT NULL	>= 0	0	Total count of streaming sessions recorded	COUNT(log_id)
total_watch_minutes	INTEGER	None	NOT NULL	>= 0	0	Sum of all session viewing durations	COALESCE(SUM(watch_duration_minutes), 0)
avg_session_minutes	REAL	None	NOT NULL	>= 0.0	0.0	Average duration per streaming session	ROUND(AVG(watch_duration_minutes), 2)
avg_completion_pct	REAL	None	NOT NULL	0.0 - 100.0	0.0	Average content completion percentage	ROUND(AVG(completion_percentage)*100, 2)
max_continuous_streak_days	INTEGER	None	NOT NULL	>= 0	0	Longest unbroken consecutive daily viewing streak	OLAP Window Function: LAG() date diff
viewer_segment	TEXT	None	NOT NULL	Segment Enum	None	Classification ('Power Streamer', 'Active Regular', 'Casual', 'Low')	CASE on total_watch_minutes
retention_status	TEXT	None	NOT NULL	Risk Enum	None	Churn vulnerability ('Retained', 'Moderate Churn Risk', 'High Risk', 'Churned')	CASE on status and watch engagement

2.10 Table / Entity: daily_platform_watchtime (Departmental Data Mart Table (SQLite))

Target Database:	viewer_watchtime_mart.db	**Key Topology:**	PK: watch_date
Business Purpose:	Time-series daily platform traffic summarizing daily active users, session frequency, and cumulative consumption hours.	**Foreign Keys:**	None (Isolated analytical mart)
Indexes / Rules:	idx_mart_daily_date ON (watch_date ASC)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
watch_date	TEXT	PK	NOT NULL	YYYY-MM-DD	None	Calendar date of platform activity	watch_logs: watch_date
active_streamers	INTEGER	None	NOT NULL	>= 0	0	Count of distinct active users streaming on that day	COUNT(DISTINCT user_id)
total_streams_logged	INTEGER	None	NOT NULL	>= 0	0	Total viewing sessions initiated during the day	COUNT(log_id)
total_watch_minutes	INTEGER	None	NOT NULL	>= 0	0	Total minutes of platform video consumed	SUM(watch_duration_minutes)
total_watch_hours	REAL	None	NOT NULL	>= 0.0	0.0	Total hours of platform video consumed	ROUND(total_watch_minutes / 60.0, 2)
avg_minutes_per_streamer	REAL	None	NOT NULL	>= 0.0	0.0	Average daily consumption per active viewer	ROUND(total_minutes / active_streamers, 2)
avg_completion_pct	REAL	None	NOT NULL	0.0 - 100.0	0.0	Average completion percentage on that day	ROUND(AVG(completion_percentage)*100, 2)

2.11 Table / Entity: watch_history_documents (Governed Document Store (MongoDB))

Target Database:	streaming_governance_dw	**Key Topology:**	PK: _id (BSON ObjectId)
Business Purpose:	Flexible semi-structured JSON document collection capturing streaming events, client device metadata, and nested genre descriptors enforced by strict $jsonSchema.	**Foreign Keys:**	user_id -> Logical, show_id -> Logical
Indexes / Rules:	MongoDB _id index, $jsonSchema validator (Strict / Error)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
session_id	String (BSON)	Unique	Required	sess_\w+	None	Unique playback session string identifier	Ingestion session UUID
user_id	Int32 (BSON)	Logical FK	Required	minimum: 1000	None	Subscriber account ID	user_subscriptions: user_id
show_id	String (BSON)	Logical FK	Required	pattern: ^s[0-9]+$	None	Catalog show identifier matching regex	content_catalog: show_id
title	String (BSON)	None	Required	minLength: 1	None	Full title string	content_catalog: title
content_type	String (BSON)	None	Required	enum: ['Movie', 'TV Show']	None	Content category enum	content_catalog: type
watch_date	String (BSON)	None	Required	pattern: ^\d{4}-\d{2}-\d{2}$	None	ISO-8601 calendar date	watch_logs: watch_date
watch_duration_minutes	Int32 (BSON)	None	Required	minimum: 1, maximum: 1440	None	Session playback length bounded by schema	watch_logs: watch_duration_minutes
completion_percentage	Double (BSON)	None	Required	minimum: 0.0, maximum: 1.0	None	Playback ratio bounded [0.0, 1.0]	watch_logs: completion_percentage
device_type	String (BSON)	None	Required	enum: ['Smart TV', 'Mobile', 'Tablet', 'Desktop', 'Gaming Console']	None	Client playback terminal device	Client telemetry payload
subscriber_tier	String (BSON)	None	Required	enum: ['Basic', 'Standard', 'Premium']	None	Subscriber service level	user_subscriptions: plan_type
genres	Array of Strings	None	Optional	items: string	[]	Nested unnested genre tags array	content_catalog: listed_in

2.12 Table / Entity: viewer_feedback_documents (Governed Document Store (MongoDB))

Target Database:	streaming_governance_dw	**Key Topology:**	PK: _id (BSON ObjectId)
Business Purpose:	Unstructured sentiment and star-rating feedback records submitted by viewers, guarded by strict $jsonSchema validators.	**Foreign Keys:**	user_id -> Logical, show_id -> Logical
Indexes / Rules:	MongoDB _id index, $jsonSchema validator (Strict / Error)	**Status:**	Active & Governed

Column / Field	Type	Key	Null	Domain / Check	Default	Business Definition	Source Lineage
feedback_id	String (BSON)	Unique	Required	fb_\w+	None	Unique feedback payload record ID	User interface submission ID
user_id	Int32 (BSON)	Logical FK	Required	minimum: 1000	None	Subscriber account ID	user_subscriptions: user_id
show_id	String (BSON)	Logical FK	Required	s\d+	None	Catalog title identifier	content_catalog: show_id
rating_score	Int32 (BSON)	None	Required	minimum: 1, maximum: 5	None	Bounded 5-star rating scale	User rating input
sentiment	String (BSON)	None	Required	enum: ['Positive', 'Neutral', 'Negative']	None	NLP sentiment classification	Automated text classifier
review_text	String (BSON)	None	Optional	maxLength: 1000	None	Optional textual viewer commentary	User commentary input
timestamp	String (BSON)	None	Required	ISO timestamp	None	Submission UTC timestamp	Server submission time

3. Strategic Indexing & Tuning Reference Matrix

To satisfy the lab requirements and eliminate costly full table scans (SCAN TABLE), twelve strategic single-column and multi-column compound B-Tree indexes were created. The table below details their physical structure and performance acceleration objectives:

Index Identifier	Target Table	Structure Type	Indexed Columns	Query Optimization Path
idx_watch_logs_user_id	watch_logs	Single B-Tree	**user_id**	Accelerates join resolutions with user_subscriptions and customer streak partitions.
idx_watch_logs_show_id	watch_logs	Single B-Tree	**show_id**	Accelerates join resolutions with content_catalog dimension table.
idx_watch_logs_watch_date	watch_logs	Single B-Tree	**watch_date**	Eliminates full table scans on temporal filtering (e.g. watch_date BETWEEN).
idx_user_subscriptions_status	user_subscriptions	Single B-Tree	**status**	Rapidly isolates 'Active' subscribers from churned/expired accounts.
idx_user_subscriptions_plan	user_subscriptions	Single B-Tree	**plan_type**	Speeds up tier-based segmentation ('Basic', 'Standard', 'Premium').
idx_content_catalog_type	content_catalog	Single B-Tree	**type**	Accelerates content filtering by 'Movie' vs. 'TV Show'.
idx_content_catalog_release_year	content_catalog	Single B-Tree	**release_year**	Enables high-speed b-tree range queries on content vintage.
idx_watch_logs_user_date	watch_logs	Compound B-Tree	**user_id, watch_date**	CRITICAL OLAP TUNING: Eliminates temporary B-Tree sorting in window functions (PARTITION BY user_id ORDER BY watch_date).
idx_watch_logs_show_dur_comp	watch_logs	Compound Covering	**show_id, watch_duration_minutes, completion_percentage**	Covering index enabling instant title aggregation without reading base table rows.
idx_watch_logs_date_comp	watch_logs	Compound B-Tree	**watch_date, completion_percentage**	Optimizes compound filters on watch date range and minimum completion ratio.
idx_user_subs_status_plan	user_subscriptions	Compound B-Tree	**status, plan_type**	Optimizes revenue attribution queries targeting active subscription tiers.
idx_catalog_type_year	content_catalog	Compound B-Tree	**type, release_year**	Powers multi-filter catalog queries combining format and release window.